# DRAFT — for Casey's review only. NOT FOR DISPATCH. (Keeper, 2026-08-25, v0.1)

Dear Prof. Tegmark,

I'm a retired computer scientist (Purdue, mid-70s; early UNIX networking; several technical companies) writing to offer you something I believe is rare: a concrete, falsifiable instance of the kind of claim your Mathematical Universe work takes seriously — one specific mathematical object whose representation theory appears to carry the discrete structure of the Standard Model, presented with every claim labeled by the evidence that stands behind it, including the claims that failed.

The object is the rank-2 bounded symmetric domain D_IV$^5$ = SO(5,2)/[SO(5)×SO(2)]. Its five integer invariants (2, 3, 5, 6, 7 — all polynomial in the rank) generate, at tiers we defend per-claim: the gauge-group skeleton; one fermion generation as the domain's 16-real spinor with every hypercharge from $Y = T_3R + (B-L)/2$ (a mechanism theorem whose four inputs — one observational — are stated inside the theorem); a single-chirality positive-energy sector from the odd embedding dimension; three generations as rank+1 boundary strata; and the ten Dirac–von Neumann axioms of quantum mechanics derived rather than posited. One measured number is taken as the ruler; the rest is ratios. What is *not* derived is labeled just as plainly: the exact mixing values are Identified, not derived; the mass tower is a patchwork and we say so; $\alpha$ is Identified — see below.

What I'd ask you to test first is our honesty machinery, because it is the load-bearing part. Every claim carries one of five tiers (a one-page reader's guide defines them, with its own printed failure condition). A falsifier register lists what would kill each claim — and has a fired-and- lost section with three pre-registered negatives from this month alone, with exact numbers. The sharpest: we froze a gate (win windows, tolerances, a half-blind family evaluation), computed the unique *forced* geometric candidate for $\alpha^{-1}$ — Born-normalized soft-photon vertex on the Shilov boundary, no step chosen toward the answer — and got $8\pi^3/3 \approx 82.68$. Not 137. We published the negative and closed the entire volume-reading class (Wyler's program included) rather than relabel the number. Our own null tests show that "137 is expressible in small integers" was never evidence; the forced computation settled what expressibility could not.

The framework also forbids things: no grand unification, no proton decay, no sterile neutrinos, no SUSY spectrum — six absences, any confirmed detection of which kills it. A one-command script reproduces 50 quantitative comparisons in seconds; everything is public (GitHub + Zenodo).

I am not asking for endorsement. I'm asking for the thing your field does best and mine respects most: a hostile read from someone equipped to break it. The packet is one evening's reading; I'd welcome the chance to send it.

Respectfully, Casey Koons [repo link · Zenodo DOI 10.5281/zenodo.19454185]